

Flex

KC Sivaramakrishnan

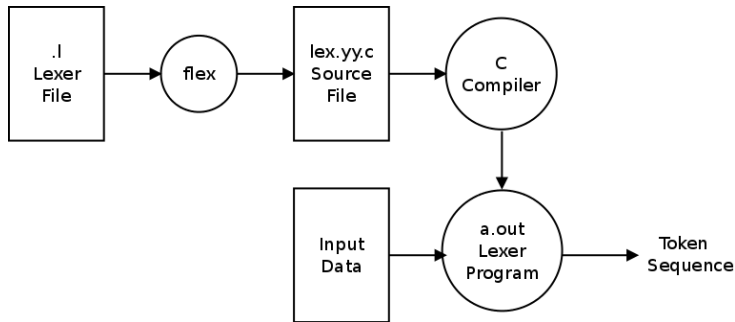
IIT Madras

Administrativa

- 6 Assignments (40% of course grade)
 - Assignment 1 : 19 points
 - Assignment 2 : 9 points
 - Assignment 3 : 19 points
 - Assignment 4 : 9 points
 - Assignment 5 : 19 points
 - Assignment 6 : 19 points
 - **Total:** 94 points
- Assignments to be done individually
 - Will use MOSS tool for plagiarism check and will run checks across previous N years.
 - 0 points + drop in grade for first instance of copying
 - U grade + DISCO for second instance.
- Mini-assignments in Lab session: 6 points
 - You may do the mini-assignments as groups of 3-4.

Flex

- Flex stands for *Fast Lexical Analyzer generator*
- Tool for generating lexical analyzers
- Manual: <https://www.cs.virginia.edu/~cr4bd/flex-manual/index.html>



Compiling a lex program

```
$ flex demo1.l  
$ gcc -o demo1.exe lex.yy.c -lfl  
$ ./demo1.exe
```

Format of the input file

definitions

%%

rules

%%

user code

Definitions section

Declarations of simple name definitions to simplify the scanner specification.

They are of the form `name definition` *unintended*.

For example,

- `DIGIT [0-9]`
- `ID [a-z][a-z0-9]*`

Rules section

Defines RE patterns and actions to take when those patterns are encountered.

They are of the form `pattern action` *unintended*.

Actions can be arbitrary C statements. If the action is empty, then the input text that matched the pattern are discarded.

Patterns are regular expressions. See <https://www.cs.virginia.edu/~cr4bd/flex-manual/Patterns.html>.

User code section

Any code copied verbatim to the output.

Values available to the user

```
char *yytext
```

holds the text of the current token. NULL terminated. It may be modified but not lengthened (you cannot append characters to the end).

```
int yyleng
```

holds the length of the current token.

Demos